1 Introduction: Emergence of Kitaev Interactions

This note will deal with a description of how Kitaev-type interactions may arise in real-world materials (magnetic insulators) via:

- Spin-Orbit Coupling (SOC).
- Crystal-field splitting.

In standard transition-metal insulators, Coulomb repulsion and the Pauli exclusion principle lead to the Heisenberg exchange term in the Hamiltonian:

$$H_{\text{Heis}} = J\vec{S}_i \cdot \vec{S}_j = J\left(S_i^x S_j^x + S_i^y S_j^y + S_i^z S_j^z\right) \tag{1}$$

Obviously, in this case every single bond interaction contains all the spin components.

To go from this to the Kitaev-type interaction, where each bond selects only one of the Cartesian spin components (i.e., $S_i^z S_j^z$, depending on bond type), we need to break $SU(2)$ spin-rotation symmetry at each site.

An obvious way for this to happen is through a spin-orbit coupling term:

$$H_{\text{SOC}} = \lambda \vec{L} \cdot \vec{S} \tag{2}$$

As this would couple the spin orientation directly with the spatial geometry of the electronic orbital on the site.

Consider a transition-metal ion such as Ru^{3+} $(4d^5)$ or Ir^{4+} $(5d^5)$. Placed in an octahedral crystal field, the 5-fold orbital degeneracy is lifted.

Side-Note: Crystal Field & Ion Environment

Note on the 5 d-orbitals: The five d-orbitals are: $d_{xy}, d_{xz}, d_{yz}, d_{x^2-y^2}, d_{3z^2-r^2}$.
Physical Environment: The crystal field is due to the ion being situated in some crystal lattice (e.g., Ru^{3+} in α-RuCl_3 or Ir^{4+} in Na_2IrO_3). There are octahedra in the unit cells, and the ligand atoms create the field that lifts the degeneracy.

2 Octahedral Crystal Field Splitting

Under an octahedral environment (O_h point group symmetry), the 5-fold orbital degeneracy of d-orbitals splits into:

1. High-energy 2-fold degenerate e_g orbitals ($d_{x^2-y^2}, d_{3z^2-r^2}$).
2. Low-energy 3-fold degenerate t_{2g} orbitals (d_{xy}, d_{xz}, d_{yz}).

The two orbitals with lobes pointing directly along the x, y, z axes (e_g) experience electrostatic repulsion from negative ligands, raising their energy. The other three orbitals (t_{2g} with lobes between axes on xy, yz, zx planes) experience reduced repulsion and lower in energy.

Let Δ_O be the octahedral crystal-field splitting parameter:

- e_g orbitals are raised by $+\frac{3}{5}\Delta_O$.
- t_{2g} orbitals are lowered by $-\frac{2}{5}\Delta_O$.

For heavy $4d$ and $5d$ ions, this Δ_O energy is larger than Hund's rule exchange energy (J_H), leading to it being favorable to place all 5 electrons in the t_{2g} manifold (low-spin state). The electronic configuration becomes:

$$(t_{2g})^5 (e_g)^0 \tag{3}$$

Since t_{2g} holds up to 6 electrons, this configuration is equivalent to a single hole moving in a filled t_{2g} background. This hole carries an effective spin $S = 1/2$.

Proof 1: Quantitative Proof of Octahedral Crystal Field Energy Shift

The total crystal field potential for an O_h symmetric ligand field can be expressed in terms of spherical harmonics $Y_{l,m}$ as:

$$V_{\text{oct}}(r,\theta,\phi) = V_0(r) + C_{40}r^4\left[Y_{4,0}(\theta,\phi) + \sqrt{\frac{5}{14}}\left(Y_{4,4}(\theta,\phi) + Y_{4,-4}(\theta,\phi)\right)\right] \tag{4}$$

Evaluating matrix elements in the d-orbital basis ($l = 2$):

$$\langle e_g|V_{\text{oct}}|e_g\rangle = +\frac{3}{5}\Delta_O, \quad \langle t_{2g}|V_{\text{oct}}|t_{2g}\rangle = -\frac{2}{5}\Delta_O \tag{5}$$

The trace conservation condition guarantees:

$$2\times\left(+\frac{3}{5}\Delta_O\right) + 3\times\left(-\frac{2}{5}\Delta_O\right) = \frac{6}{5}\Delta_O - \frac{6}{5}\Delta_O = 0 \tag{6}$$

3 Effective Angular Momentum and the T-P Isomorphism

We restrict the atomic angular momentum operator $\vec{L}$ ($L = 2$) to the 3D t_{2g} subspace $\{|d_{xy}\rangle, |d_{xz}\rangle, |d_{yz}\rangle\}$.

In this basis, we can derive the matrix representations of the orbital angular momentum operators:

$$(L_x)_{t_{2g}} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & i \\ 0 & -i & 0 \end{pmatrix} \tag{7}$$

$$(L_y)_{t_{2g}} = \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix} \tag{8}$$

$$(L_z)_{t_{2g}} = \begin{pmatrix} 0 & i & 0 \\ -i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \tag{9}$$

Now we consider an effective $l_{\text{eff}} = 1$ orbital momentum operator $\vec{l}_{\text{eff}}$ acting in a 3D spherical basis. If we define the following states in the t_{2g} basis:

$$|l_z = 0\rangle \equiv |d_{xy}\rangle \tag{10}$$

$$|l_z = +1\rangle \equiv -\frac{1}{\sqrt{2}}\left(i|d_{xz}\rangle + |d_{yz}\rangle\right) \tag{11}$$

$$|l_z = -1\rangle \equiv \frac{1}{\sqrt{2}}\left(i|d_{xz}\rangle - |d_{yz}\rangle\right) \tag{12}$$

We can directly see that:

$$\vec{L}_{t_{2g}} = -\vec{l}_{\text{eff}} \tag{13}$$

The negative sign is the consequence of group theory, called the **T-P isomorphism** (T-P equivalence). The t_{2g} manifold behaves like an effective $l_{\text{eff}} = 1$ orbital angular momentum.

Proof 2: Explicit Derivation of Matrix Projections & T-P Isomorphism

Expressing the d-orbitals in terms of spherical harmonics $Y_{2,m}$:

$$|d_{xy}\rangle = \frac{-i}{\sqrt{2}}(|2,2\rangle - |2,-2\rangle), \quad |d_{xz}\rangle = \frac{-1}{\sqrt{2}}(|2,1\rangle - |2,-1\rangle), \quad |d_{yz}\rangle = \frac{i}{\sqrt{2}}(|2,1\rangle + |2,-1\rangle) \tag{14}$$

Applying $L_z = -i(x\partial_y - y\partial_x)$:

$$L_z|d_{xz}\rangle = L_z\left[-\frac{1}{\sqrt{2}}(|2,1\rangle - |2,-1\rangle)\right] = -\frac{1}{\sqrt{2}}(1|2,1\rangle - (-1)|2,-1\rangle) = i|d_{yz}\rangle \tag{15}$$

$$L_z|d_{yz}\rangle = L_z\left[\frac{i}{\sqrt{2}}(|2,1\rangle + |2,-1\rangle)\right] = \frac{i}{\sqrt{2}}(1|2,1\rangle + (-1)|2,-1\rangle) = -i|d_{xz}\rangle \tag{16}$$

$$L_z|d_{xy}\rangle = 2i|d_{x^2-y^2}\rangle \xrightarrow{\text{projection onto } t_{2g}} 0 \tag{17}$$

Writing L_z in basis $\{|d_{xy}\rangle, |d_{xz}\rangle, |d_{yz}\rangle\}$ yields:

$$(L_z)_{t_{2g}} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & i \\ 0 & -i & 0 \end{pmatrix} \tag{18}$$

Comparing $(L_z)_{t_{2g}}$ with standard $l = 1$ spin-1 matrices $(l_z)_{m,m'} = m\delta_{m,m'}$ evaluated in the transformed basis $\{|+1\rangle, |0\rangle, |-1\rangle\}$ confirms:

$$\langle\psi_a|\vec{L}|\psi_b\rangle = -\langle\phi_a|\vec{l}_{\text{eff}}|\phi_b\rangle \implies \vec{L}_{t_{2g}} = -\vec{l}_{\text{eff}} \tag{19}$$

4 Spin-Orbit Coupling and the $J_{\text{eff}} = 1/2$ Kramers Doublet

For $4d$ and $5d$ transition elements, the spin-orbit coupling term is large. The SOC Hamiltonian for the single t_{2g} hole with effective spin $S = 1/2$ becomes:

$$H_{\text{SOC}} = \lambda\vec{L}\cdot\vec{S} = -\lambda\vec{l}_{\text{eff}}\cdot\vec{S} \tag{20}$$

By shifting to a suitable basis or combining angular momenta on the 6-dimensional space (3 orbital $\times$ 2 spin), we get sectors characterized by total effective angular momentum $\vec{J}_{\text{eff}} = \vec{l}_{\text{eff}} + \vec{S}$:

- $J_{\text{eff}} = 3/2$ (4-fold degenerate quartet)
- $J_{\text{eff}} = 1/2$ (2-fold degenerate doublet)

We can calculate the energy levels:

1. For $J_{\text{eff}} = 3/2$, the energy level is $-\frac{1}{2}\lambda$ (or $+\frac{1}{2}\lambda$ for electrons).
2. For $J_{\text{eff}} = 1/2$, the energy level is $+\lambda$ (or $-\lambda$ for electrons).

The $J_{\text{eff}} = 1/2$ doublet lies lowest in energy for holes, owing to the negative sign from the T-P isomorphism step. The ground state doublet is separated from the excited quartet by an energy gap of:

$$\Delta_{\text{SOC}} = \frac{3}{2}\lambda \tag{21}$$

Math Verification: Electron vs. Hole Representation Sign Convention

Let us explicitly verify the energy eigenvalues of $\vec{l}_{\text{eff}} \cdot \vec{S}$:

$$\vec{l}_{\text{eff}} \cdot \vec{S} = \frac{1}{2}\left(\vec{J}^2_{\text{eff}} - \vec{l}^2_{\text{eff}} - \vec{S}^2\right) = \frac{1}{2}\left[J_{\text{eff}}(J_{\text{eff}}+1) - 1(2) - \frac{3}{4}\right] = \frac{1}{2}\left[J_{\text{eff}}(J_{\text{eff}}+1) - \frac{11}{4}\right] \quad (22)$$

- For $J_{\text{eff}} = 3/2$: $\vec{l}_{\text{eff}} \cdot \vec{S} = \frac{1}{2}\left(\frac{15}{4} - \frac{11}{4}\right) = +\frac{1}{2}$. Thus, $H_{\text{SOC}} = -\lambda(+1/2) = -\frac{1}{2}\lambda$.
- For $J_{\text{eff}} = 1/2$: $\vec{l}_{\text{eff}} \cdot \vec{S} = \frac{1}{2}\left(\frac{3}{4} - \frac{11}{4}\right) = -1$. Thus, $H_{\text{SOC}} = -\lambda(-1) = +\lambda$.

Hole vs. Electron Correction: For electrons, $\lambda > 0$ means $J_{\text{eff}} = 3/2$ is lowest. However, for a d^5 configuration (5 electrons $\equiv$ 1 hole in t_{2g} shell), the hole spin-orbit constant has an inverted sign ($\lambda_{\text{hole}} = -\lambda_{\text{electron}}$). Therefore, for holes, the $J_{\text{eff}} = 1/2$ doublet forms the true single-hole ground state.

5 Single-Ion Hamiltonian with Non-Cubic Distortion

In real crystal lattices, a compression or elongation along the trigonal/tetragonal z-axis introduces a non-cubic crystal field distortion term, $\Delta(l_z^{\text{eff}})^2$.

The single-ion Hamiltonian then becomes:

$$H_0 = -\lambda \vec{l}_{\text{eff}} \cdot \vec{S} + \Delta(l_z^{\text{eff}})^2 \quad (23)$$

In the basis $\{|J_{\text{eff}}, m_j\rangle\}$ consisting of states $\{|3/2, \pm 3/2\rangle, |3/2, \pm 1/2\rangle, |1/2, \pm 1/2\rangle\}$, the single-ion matrix takes the form:

$$H_0 = \begin{pmatrix} \Delta_+ & 0 & 0 & 0 & 0 & 0 \\ 0 & \Delta_- & 0 & 0 & \sqrt{2}\lambda & 0 \\ 0 & 0 & \Delta_- & 0 & 0 & \sqrt{2}\lambda \\ 0 & 0 & 0 & \Delta_+ & 0 & 0 \\ 0 & \sqrt{2}\lambda & 0 & 0 & \lambda + \frac{\Delta}{3} & 0 \\ 0 & 0 & \sqrt{2}\lambda & 0 & 0 & \lambda + \frac{\Delta}{3} \end{pmatrix} \quad (24)$$

where $\Delta_\pm = \Delta \pm \frac{\lambda}{2}$.

Diagonalizing this Hamiltonian yields the explicit ground-state wavefunction for the $J_{\text{eff}} = 1/2$ pseudo-spin doublet:

$$|\bar{\uparrow}\rangle = \sin\theta|0, \uparrow\rangle - \cos\theta|+1, \downarrow\rangle \quad (25)$$

$$|\bar{\downarrow}\rangle = \sin\theta|0, \downarrow\rangle - \cos\theta|-1, \uparrow\rangle \quad (26)$$

where the mixing angle θ is defined by:

$$\tan 2\theta = \frac{\sqrt{2}\lambda}{\frac{\lambda}{2} - \Delta} \quad (27)$$

This doublet forms the fundamental local pseudo-spin operator $\vec{S}_i$ at each lattice site.

6 Inter-Site Exchange Geometry

In typical transition-metal magnetic insulators, there is barely any direct TM d-orbital overlap due to them being separated by negative ligand ions (O^{2-}, Cl^-, etc.). A virtual electron can thus hop

from one TM ion to another through an intermediate ligand. Let us call this a **superexchange path**.

Now, there are 2 main geometric types of TM-ligand-TM bonds:

1. 180° **straight bond (corner-sharing):** Only one path of this type exists between neighboring TM ions.
2. 90° **bond (edge-sharing):** There are 2 parallel paths of this type linking adjacent octahedra.

The corner-sharing path connects identical d-orbitals on both sides (i.e., $d_{xy}^{(i)} \leftrightarrow d_{xy}^{(j)}$).

Kramers-Anderson superexchange theory allows us to calculate the hopping amplitudes on each path. In general, the detailed calculation is in-depth, but the key results are:

- Hopping through the 180° corner-sharing path yields standard isotropic Heisenberg exchange in terms of pseudo-spin:
$$J\vec{S}_i \cdot \vec{S}_j \tag{28}$$
- Along with a dipolar anisotropy term:
$$J_2(\vec{S}_i \cdot \hat{r}_{ij})(\hat{r}_{ij} \cdot \vec{S}_j) \tag{29}$$

7 Cancellation of Heisenberg Exchange in Edge-Sharing Geometry

The competing 90° edge-sharing paths destructively interfere with each other. This is due to the intermediate ligand p-orbitals linking orthogonal TM d-orbitals (e.g., $d_{yz}^{(i)}$ and $d_{xz}^{(j)}$) and the $\pi/2$ rotation symmetry around the principal axis.

As a result, the lowest-order isotropic Heisenberg exchange term that normally dominates superexchange vanishes here in an ideal 90° edge-sharing geometry . Second-order virtual processes involving Hund's rule exchange (J_H) on the ligands contribute to anisotropic terms, yielding no dominant Heisenberg interaction.

With this established, since each edge-sharing bond plane only couples specific components of orbital motion, we obtain the mapping summary presented in Table 1.

Table 1: Bond Geometry, TM-O Planes, and Anisotropic Coupling Types

Bond Type (α)	**TM-O Plane**	**Perpendicular Axis**	**Coupling Term**
z-bond	xy-plane	z-axis	$-KS_i^zS_j^z$
x-bond	yz-plane	x-axis	$-KS_i^xS_j^x$
y-bond	zx-plane	y-axis	$-KS_i^yS_j^y$

Which means that the Hamiltonian on an α-type bond has the bond-directional anisotropy:

$$H_{\langle ij\rangle_\alpha} = -KS_i^\alpha S_j^\alpha \tag{30}$$

Here, the Kitaev coupling strength is given perturbatively by:

$$K \approx \frac{8t^2 J_H}{3U^2} > 0 \tag{31}$$

where t is the d-p hopping integral, J_H is Hund's coupling, and U is the on-site Coulomb repulsion.

8 The Complete Kitaev-Heisenberg-Γ Model

Combining the isotropic exchange J and the anisotropic Kitaev coupling K, we recover the full Kitaev-Heisenberg (KH) Hamiltonian:

$$H_{\mathsf{KH}} = -K \sum_{\gamma\in\{x,y,z\}} \sum_{\langle ij\rangle_\gamma} S_i^\gamma S_j^\gamma + J \sum_{\langle ij\rangle} \vec{S}_i \cdot \vec{S}_j \tag{32}$$

Breaking of ideal 90° edge-sharing geometry in real materials (e.g., lattice distortion, direct d-d overlap, and SOC-hopping interference) introduces an additional symmetric off-diagonal exchange term Γ:

$$H_\Gamma = \Gamma \sum_{\gamma\in\{x,y,z\}} \sum_{\langle ij\rangle_\gamma} \left(S_i^{\gamma'} S_j^{\gamma''} + S_i^{\gamma''} S_j^{\gamma'} \right) \tag{33}$$

where $\gamma' \neq \gamma'' \neq \gamma$ denote the two Cartesian directions orthogonal to bond direction γ.

With this term included, the model is called the **Kitaev-Heisenberg-Γ (KHΓ) model**.

Example: On a z-type bond ($\gamma = z$, so $\gamma' = x, \gamma'' = y$), the explicit KH$\Gamma$ interaction is:

$$H_{\langle ij\rangle_z} = -K S_i^z S_j^z + J \vec{S}_i \cdot \vec{S}_j + \Gamma \left(S_i^x S_j^y + S_i^y S_j^x \right) \tag{34}$$

All in all, this microscopic derivation of bond-directional exchange interactions in d^5 spin-orbit coupled insulators is referred to as the **Jackeli-Khaliullin mechanism**.